FIRST REGISTRARS & INVESTOR SERVICES LTD

RFID ASSET MANAGEMENT SYSTEM

USER TRAINING MANUAL (MVP VERSION)

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Version: 1.0

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CHAPTER 1 - INTRODUCTION

Purpose

The RFID Asset Management System enables First Registrars & Investor Services Ltd to electronically manage, track and verify all company assets using RFID technology. The system improves accountability, simplifies asset audits, reduces losses and provides real-time visibility of company assets throughout their lifecycle.

Objectives

At the end of this training, users should be able to:

- Navigate the application
- Create users (Back Office)
- Register assets
- Assign RFID tags
- Verify assets using handheld RFID readers
- Assign assets to employees
- Transfer assets between locations
- Conduct RFID stock verification
- Generate reports
- Maintain complete asset histories

CHAPTER 2 - USER ROLES

Staff

Responsibilities:

- View assigned assets
- View asset details
- Request asset transfers
- Report damaged or lost assets

Back Office

Responsible for daily asset operations. Functions:

- Create and manage users
- Maintain departments and locations
- Register assets
- Assign RFID tags
- Assign assets to staff
- Process transfers
- Conduct stock verification
- Generate operational reports

Auditor

Responsible for:

- Asset verification
- Audit reports
- Compliance reviews
- Read-only access to operational records

Administrator

Responsible for:

- System configuration
- Security settings
- Role and permission management
- RFID reader configuration
- Backups and recovery
- System maintenance

CHAPTER 3 - LOGIN

Login URL: http://localhost:5000/Account/Login

Login Steps

- Enter Username
- Enter Password
- Click Login
- Change the default password on first login (if prompted)

Default Demo Accounts

The following accounts are pre-configured in the system for training and testing. Use the username and password shown to access the corresponding portal.

Username	Password	Role	Portal
Washington	Washington1	Backoffice	Back Office
Staff	Staff1	Staff	Staff
Auditor	auditor1	Auditor	Auditor
Admin	Admin1	Admin	Administrator
oludele.gbenro@firstregistrarsnigeria.com	Password@123	Backoffice	Back Office
Staff1	January2021###	Staff	Staff
Staff2	January2021###	Staff	Staff
Staff3	January2021###	Staff	Staff
Staff4	January2021###	Staff	Staff
staff	staff123	Staff	Staff
viewer	viewer123	Viewer	Read-only (limited access)

Recommended Training Accounts

For the practical assessment in Chapter 20, use these primary accounts:

- Back Office operations: Washington / Washington1
- Staff self-service: Staff / Staff1
- Audit and verification: Auditor / auditor1
- System administration: Admin / Admin1

CHAPTER 4 - DASHBOARD

Each dashboard provides role-specific information.

Back Office Dashboard

- Total Assets
- Assets Assigned
- Assets Awaiting RFID
- Assets Without Custodian
- Pending Transfers
- Recent Activities

Auditor Dashboard

- Verification Progress
- Audit Exceptions
- Missing Assets

Staff Dashboard

- My Assigned Assets
- Pending Requests

Administrator Dashboard

- Active Users
- RFID Readers
- System Health
- Audit Logs

CHAPTER 5 - MASTER DATA SETUP

Before registering assets, configure:

- Branches
- Buildings
- Floors
- Rooms
- Departments
- Asset Categories
- Asset Types
- Vendors
- Manufacturers

These records ensure consistent data entry throughout the system.

CHAPTER 6 - USER MANAGEMENT (BACK OFFICE)

Create a User

Navigate to: Administration -> Users -> Create User

Complete:

- Employee ID
- Name
- Email
- Department
- Branch
- Designation
- Phone Number
- User Role
- Temporary Password

Click Save.

Manage Users

Back Office can:

- Edit user details
- Activate or deactivate users
- Reset passwords
- Search by Employee ID, Name or Department

CHAPTER 7 - ASSET REGISTRATION

To register a new asset:

- Go to Assets -> New Asset

Enter:

- Asset Number
- Category
- Asset Type
- Description
- Manufacturer
- Model
- Serial Number
- Purchase Date
- Purchase Cost
- Vendor
- Department
- Default Location
- Save the asset.

The asset is now recorded but is NOT yet RFID-enabled.

CHAPTER 8 - RFID TAG MANAGEMENT

RFID tags must be unique. The system supports:

- Assign RFID Tag
- Replace RFID Tag
- Remove RFID Tag
- Verify RFID Tag

The system prevents duplicate RFID assignments.

RFID Tag Statuses

- Blank
- Assigned
- Active
- Damaged
- Lost
- Retired

CHAPTER 9 - EXISTING ASSET RFID MIGRATION

This process is used for assets that already exist in the database.

Workflow

- Search for the asset.
- Open the asset record.
- Click Assign RFID Tag.
- Connect the RFID handheld reader.
- Select Write Tag.
- Place a blank RFID tag on the reader.
- The system writes a unique EPC to the tag.
- The EPC is linked permanently to the selected asset.
- Verify the tag by scanning it.
- Save.

The asset is now RFID-enabled.

CHAPTER 10 - REGISTERING NEW RFID ASSETS

- Register the asset.
- Save the asset record.
- Select Assign RFID Tag.
- Encode a blank RFID tag.
- Verify the RFID tag.
- Mark the asset as Available.

CHAPTER 11 - USING THE RFID HANDHELD READER

The RFID reader has three primary operating modes.

Read Mode

Purpose: Identify an existing RFID-tagged asset.

App screen: RFID Scan -> Read Mode (/RfidReader?mode=read)

Steps:

- Open RFID Scan from the portal sidebar.
- Select Read Mode.
- Tap the RFID Code field.
- Scan the tag with the handheld reader (or type the code).
- Submit - asset details appear automatically.

Write Mode

Purpose: Encode a new RFID tag and link it to an asset.

App screen: RFID Scan -> Write Mode (/RfidReader?mode=write) or Assets -> Assign RFID Tag

Steps:

- Select the asset in the system.
- Encode the blank tag using Zebra 123RFID Mobile (hardware) or Write RFID Tag in the app.
- Place a blank RFID tag on the reader.
- Wait for confirmation.
- Verify by scanning the newly encoded tag in Verification Mode.

Verification Mode

Purpose: Confirm that an RFID tag is correctly linked to the asset.

App screen: RFID Scan -> Verification Mode (/RfidReader?mode=verify)

The system displays:

- Asset Number
- Description
- Current Custodian
- Department
- Current Location
- Status

Unknown RFID Tags

If the reader detects an unknown RFID tag, the system displays:

Unknown RFID Tag Detected

Options:

- Register as New Asset
- Link to Existing Asset
- Ignore

CHAPTER 12 - ASSET ASSIGNMENT

- Open the asset record.
- Click Assign Asset.
- Select the employee.
- Scan the RFID tag for verification.
- Confirm the assignment.

The employee becomes the current custodian.

CHAPTER 13 - ASSET TRANSFER

- Select the asset.
- Choose the destination department or location.
- Submit the transfer request.
- Upon approval, scan the asset before it leaves its current location.
- Scan the asset again upon arrival.
- Complete the transfer.

The movement history is recorded automatically.

CHAPTER 14 - RFID STOCK VERIFICATION

Purpose: Verify the physical presence of assets.

Steps:

- Select a department or room.
- Start Stock Verification.
- Walk through the area with the RFID reader.
- The reader automatically detects tagged assets.
- The system compares scanned assets with expected records.

Results

- Found Assets
- Missing Assets
- Unexpected Assets
- Duplicate Reads

Generate the verification report after completion.

CHAPTER 15 - SEARCH

Search by:

- Asset Number
- RFID Tag
- Serial Number
- Employee
- Department
- Location
- Manufacturer

CHAPTER 16 - REPORTS

Available reports include:

- Asset Register
- Employee Asset Register
- Department Asset Report
- RFID Assignment Report
- Assets Without RFID
- Verification Report
- Missing Assets
- Asset Movement Report
- Audit Report

Reports can be exported to Excel or PDF.

CHAPTER 17 - AUDIT TRAIL

Every action performed in the system is recorded.

Captured information includes:

- User
- Action
- Date
- Time
- Previous Value
- New Value

Audit records cannot be modified or deleted.

CHAPTER 18 - BEST PRACTICES

- Never share login credentials.
- Verify an RFID tag immediately after encoding.
- Scan assets before and after every transfer.
- Investigate unknown RFID tags promptly.
- Do not reuse retired RFID tags without an approved process.

- Perform regular stock verification.

CHAPTER 19 - TROUBLESHOOTING

Problem: RFID tag not detected.

Action:

- Confirm the reader is connected.
- Ensure the tag is within reading range.
- Verify the tag has been encoded.

Problem: Duplicate RFID detected.

Action:

- Locate the existing asset linked to the tag.
- Do not assign the same RFID tag to another asset.

Problem: User cannot log in.

Action:

- Verify the account is active.
- Reset the password if required.

Problem: Zebra RFD40 scan not appearing in browser.

Action:

- Confirm the RFID input field is focused before scanning.
- Verify DataWedge profile is associated with Chrome browser.
- Check RFID Input is enabled and RFD40 is selected as reader.
- Update RFD40 firmware to 006-R01 or higher.
- Re-pair sled using Zebra Bluetooth Pairing Utility if using Bluetooth.

CHAPTER 20 - PRACTICAL ASSESSMENT

Each trainee must complete the following end-to-end exercise:

- Create a new staff user.
- Register a new laptop asset.
- Encode and assign an RFID tag.
- Verify the tag using the handheld reader.
- Search for an existing asset and assign it an RFID tag.
- Assign the new laptop to the staff user.
- Transfer the laptop to another department.
- Conduct an RFID stock verification.
- Generate an Asset Register report.
- Review the audit trail for all actions performed.

A trainee is considered competent when they can complete the full workflow independently and accurately.

CHAPTER 21 - RECOMMENDED RFID READERS

Use UHF passive RFID tags (EPC Gen2 / ISO 18000-6C) for all asset labels. The system supports readers that output tag data via keyboard wedge (DataWedge) or via REST API integration.

Handheld UHF Readers (tagging, verification, stock take)

- Zebra RFD40 (with TC21/TC26/TC52/TC53 mobile computer) - recommended for enterprise rollout
- Zebra RFD90 - longer range, rugged environments
- Honeywell IH25 - solid handheld for inventory audits
- Chainway C72 / C5 - cost-effective Android handheld with built-in UHF
- TSL 1128 - Bluetooth UHF sled used with phone or tablet

Fixed Door / Gate Readers (exit monitoring)

- Impinj Speedway R420 / R700 with antennas - industry standard portal readers
- Zebra FX7500 / FX9600 - fixed reader for doorways and choke points
- ThingMagic Sargas - compact fixed reader for single-door setups

Desktop USB Readers (encoding stations)

- ThingMagic USB Plus+ - desktop UHF for back-office tag encoding
- Generic UHF USB readers - suitable for low-volume tagging desks

Tag Types

- Standard adhesive UHF labels - furniture, general equipment
- On-metal UHF tags - laptops, servers, metal cabinets
- Rugged high-memory tags - vehicles and large outdoor assets

Suggested Rollout for First Registrars

- Phase 1 (Training): Chainway C72 or Zebra RFD40 + TC26 in keyboard wedge mode
- Phase 2 (Operations): Zebra RFD40 fleet with DataWedge profiles pushed via StageNow
- Phase 3 (Security): Zebra FX9600 at main entrance with API middleware for exit alerts

CHAPTER 22 - CONNECTING READERS TO FRISL EAMS

FRISL EAMS is a web application. RFID readers do not plug into the server directly. They connect to a client device (PC, tablet, or Zebra mobile computer) that runs the browser or a middleware service posting scans to the API.

Connection Methods

Method 1 - Keyboard Wedge / DataWedge (recommended for handhelds)

- Reader sends EPC as keystrokes into the focused browser field.
- Works today with no custom development.
- Use with /RfidReader, /StockVerification, /Assignments, /Transfers.

Method 2 - REST API (recommended for door readers and batch stock take)

- Middleware or custom app POSTs scan results to FRISL API endpoints.
- Suitable for fixed portal readers and high-volume stock verification.
- See Chapter 24 for API details.

Method 3 - Direct USB/Bluetooth SDK in browser

- Not supported in the current MVP web application.

App Screens for RFID Workflows

- Read / verify tag: /RfidReader?mode=read or verify
- Write / assign tag: /RfidReader?mode=write or /RfidTags/Assign
- Stock verification: /StockVerification
- Assignment RFID confirm: /Assignments
- Transfer departure / arrival: /Transfers
- Exit / door monitoring: /Workflow/Rfid

Quick Test Without Hardware

Open /RfidReader?mode=verify, type a seeded tag such as RFID-000101, and submit. Asset details should appear if the tag is linked in the system.

CHAPTER 23 - ZEBRA RFD40 + TC-SERIES SETUP GUIDE

This chapter describes how to connect the Zebra RFD40 UHF sled with a TC-series mobile computer (TC21, TC26, TC52, TC53, TC58) to FRISL EAMS.

Hardware Required

- Zebra RFD40 sled (Standard, Premium, or Premium Plus)
- Zebra TC21/TC26/TC52/TC53/TC58 mobile computer
- eConnex adapter (e.g. ADP-RFD40-TC2X-1E) for snap-on connection - recommended
- OR Bluetooth adapter (ADP-RFD40-TC2X-1R) for wireless connection
- UHF EPC Gen2 asset tags (on-metal tags for IT equipment)
- Note: eConnex requires TC device with 8-pin rear connector SKU

Physical Connection - eConnex (Recommended)

- Attach RFD40 to the TC mobile computer using the correct eConnex adapter.
- Power on both the sled and the mobile computer.
- The sled is detected via the 8-pin connector (Bluetooth is disabled in this mode).
- Connect TC to Wi-Fi or mobile data.
- Open Chrome and navigate to the FRISL login URL.

Physical Connection - Bluetooth

- Attach the Bluetooth adapter to the RFD40.
- Open Zebra Bluetooth Pairing Utility on the TC device.
- Scan the pairing barcode and tap PAIR.
- Configure DataWedge mode (see below).

Software Prerequisites

- RFD40 firmware version 006-R01 or higher
- DataWedge version 13.0 or higher on the TC device
- 123RFID Desktop (PC) or 123RFID Mobile (TC) for initial configuration

Configure RFD40 for DataWedge Mode

- Connect RFD40 to PC via USB or use 123RFID Mobile on the TC.
- Set regulatory region (Country of Operation) for your location.
- Set Bluetooth Host Type = DataWedge (for Bluetooth setups).
- Pair sled to TC using Bluetooth Pairing Utility if applicable.
- On TC: open DataWedge -> create profile for Chrome browser.
- In profile: enable RFID Input, select RFD40 as reader.
- Set output to Keystroke (keyboard wedge). Optionally suffix with Enter after scan.

DataWedge Profile Checklist

- Associated app: Chrome (or Zebra Enterprise Browser)
- RFID Input: Enabled
- Reader selection: RFD40
- Output: Keystroke

- Key event: Enter after scan (optional, for auto-submit workflows)

Using RFD40 with FRISL EAMS

Login URL: http://<server>:5000/Account/Login

Recommended training account: Washington / Washington1 (Back Office)

Workflow steps on TC device:

- Sign in to FRISL EAMS in Chrome.
- Open the required screen (e.g. RFID Scan -> Verification Mode).
- Tap the RFID Code input field to focus it.
- Pull the RFD40 trigger and scan the asset tag.
- The EPC appears in the field - press Submit or Enter.

Workflow Mapping

- Read / verify: /RfidReader?mode=verify - tap field, scan, submit
- Stock verification: /StockVerification - start session, scan each tag
- Assignment confirm: /Assignments - scan into verification field
- Transfer scans: /Transfers - scan at departure and arrival
- Assign tag to asset: Asset detail -> Assign RFID Tag - scan EPC

Write Mode (Encode New Tags) - Important

Real EPC encoding to blank RFID tags requires Zebra 123RFID Mobile or the Zebra RFID SDK on the TC device. The FRISL web app Write Mode simulates linking for training. Production workflow: (1) encode tag with Zebra tools, (2) link EPC in FRISL via Assign RFID Tag.

Architecture Overview

```
RFD40 sled --(DataWedge keystroke)--> TC26 Chrome --> FRISL EAMS
                         /RfidReader, /StockVerification, /Assignments
```

Zebra RFD40 Troubleshooting

- No scan in browser: confirm RFID field is focused; check DataWedge profile for Chrome
- Four beeps on scan: DataWedge decode error - re-pair sled, check firmware
- RFD40 not in DataWedge reader list: update firmware to 006-R01+; Premium Plus required for RFID DataWedge
- eConnex not detected: verify correct adapter SKU and 8-pin TC rear connector
- Tag not found in app: tag not linked - use Assign RFID Tag in FRISL

CHAPTER 24 - RFID API INTEGRATION (PRODUCTION)

For fixed door readers, batch stock take, or custom Zebra apps, POST scan data to the FRISL REST API. API documentation is available at /swagger in development.

Single Door / Portal Scan

POST /api/monitoring/rfid-scan

```
{
  "rfidCode": "RFID-000101",
  "doorLocation": "Main Gate"
}
```

Used for exit monitoring and door alerts (/Workflow/Rfid).

Batch Scans (Stock Take / Handheld Upload)

POST /api/integration/rfid-batch

```
{
  "sourceSystem": "Zebra-RFD40-TC26",
  "scans": [
    { "rfidCode": "RFID-000101", "doorLocation": "IT Department" },
    { "rfidCode": "RFID-000102", "doorLocation": "IT Department" }
  ]
}
```

A small middleware service or Zebra RFID SDK app on the TC can collect scans during a stock walk and upload them in one batch.

Fixed Reader Middleware Architecture

```
RFID antenna -> Fixed reader (FX9600/Impinj)
            -> Middleware service
            -> POST /api/monitoring/rfid-scan
            -> FRISL exit monitoring & alerts
```

Connection Method Summary

- Keyboard wedge / DataWedge: Yes - works with web forms today (low effort)
- REST API: Yes - for doors and batch uploads (medium effort)
- Direct browser SDK: Not built in MVP (high effort)
- Native mobile app with Zebra SDK: Optional for production batch scans